Andhra University College of Engineering

Computer Science and Engineering | Course Code: CS302

Database Management Systems - Laboratory Manual & Practical Guide

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Syllabus Focus: CS302 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Relational Data Model & Formal Languages

" Syllabus Scope: Database system architectures, Entity-Relationship ER modeling, Relational algebra oper

Unit 2: Relational Query Language SQL

" Syllabus Scope: DDL, DML, DCL commands, Nested subqueries, Correlated subqueries, Aggregate functions,

CS302: Database Management Systems - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Relational Database Design & Normalization

" Functional dependencies, Armstrong axioms, Attribute closure, 1NF, 2NF, 3NF, Boyce-Codd Normal Form BCNF, 4N

Unit 4: Transaction Processing & Concurrency Control

" ACID properties, Serializability, Conflict vs View serializability, Two-Phase Locking 2PL, Strict 2PL, Times

Unit 5: Storage, Indexing & Query Processing

" RAID levels, B-Trees, B+ Trees node splitting and deletion, Static and Dynamic hashing, Cost-based query exe

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Database Schema Definition, Table Constraints, and Complex SQL Query Execution in PostgreSQL

Lab 2: Stored Procedures, Functions, Cursors, and Database Event Triggers

Lab 3: Normalization Case Study: Redesigning an Unnormalized Enterprise Schema to BCNF

Lab 4: Transaction Concurrency Testing: Simulating Dirty Reads, Non-Repeatable Reads, and Deadlocks

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Raghu Ramakrishnan, Johannes Gehrke - Database Management Systems McGraw-Hill 3rd Ed

* Abraham Silberschatz, Henry F. Korth - Database System Concepts McGraw-Hill 7th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students